

UNIT CODE	HF2J
UNIT NAME	WEB COMPONENT DEVELOPMENT I
ASSESSMENT TYPE	WEB APPLICATION DEVELOPMENT PROJECT, RECORDED SOLUTION DEMONSTRATION, AND LIVE VIVA VOCE EXAMINATION

ASSESSMENT GUIDE

OVERVIEW

This assessment evaluates the student's ability to analyse, design, develop, deploy, test, explain, and defend a Java EE Web Application based on a given business scenario.

Students are required to:

- Develop a complete Java EE Web Application.
- Submit a recorded solution demonstration.
- Attend a Live Viva Voce Examination and Project Defence.

The assessment is designed to evaluate both theoretical understanding and practical Web Component Development skills while verifying the authenticity and ownership of submitted work.

OVERALL ASSESSMENT WEIGHTING

COMPONENT	MARKS
Component A – Web Application Development Project	50
Component B – Recorded Solution Demonstration	20
Component C – Live Viva Voce Examination and Project Defence	30
TOTAL	**100 MARKS**

IMPORTANT NOTE

Assessment marks will be awarded based on the student's ability to:

- Develop an appropriate web application solution.
- Apply Servlet, JSP, Session Management, Security, and Web Container concepts.
- Analyse requirements and technical design decisions.
- Configure deployment descriptors correctly.
- Implement session tracking and security mechanisms.
- Demonstrate successful deployment.
- Explain, justify, and defend submitted work.
- Demonstrate ownership and authenticity throughout the assessment process.

SCENARIO

SkillHub Training Academy is a rapidly expanding professional training organisation that delivers technology, business, and professional development programmes.

The organisation currently manages student registrations, course enrolments, trainer information, attendance records and manual administration processes.

Management has decided to develop a web-based training management portal to improve operational efficiency, automate learner services, and provide secure online access to training information.

Students are required to analyse the requirements and develop a suitable Java EE web application solution using only Java EE scopes and in-memory data structures, without using a database management system.

Reasonable assumptions may be made where necessary. All assumptions must be clearly documented.

COMPONENT A – WEB APPLICATION DEVELOPMENT PROJECT

OVERVIEW

Students are required to develop a complete Java EE Web Application for the given scenario.

DEVELOPMENT REQUIREMENTS

TASK	REQUIREMENTS
Requirements Analysis	Identify users, functions, business requirements, assumptions, and constraints.
Application Planning	Explain system purpose, scope, users, objectives, and expected benefits.
Web Application Structure	Design application architecture and directory structure.
Deployment Configuration	Configure deployment descriptors and application settings.
Servlet Development	Develop servlets for request processing and business logic.
JSP Development	Develop JSP pages for user interaction and presentation.
Web Container Features	Implement ServletContext, RequestDispatcher, Filters, Listeners, and Attribute Scopes.
Session Management	Implement login sessions and session tracking mechanisms.
Security Implementation	Implement authentication, authorisation, and security configurations.
Deployment and Testing	Deploy the application and test functionality.
Operational Maintenance	Briefly explain maintenance, security, scalability, and future improvements.

MINIMUM APPLICATION REQUIREMENTS

The developed application shall include:

- Minimum 5 Servlets
- Minimum 6 JSP Pages
- Login Functionality
- Session Management
- RequestDispatcher Usage
- At least 1 Listener

- web.xml Configuration
- Authentication and Authorisation Mechanism
- Logout Functionality
- WAR Deployment

SUBMISSION FORMAT

SECTION	CONTENT REQUIREMENTS
Cover Page	Project Title, Student Name, Student ID, Batch, Unit Name, Submission Date
1. Introduction	Overview of the scenario and proposed solution
2. Requirements Analysis	Users, requirements, assumptions, constraints
3. Application Planning	Objectives, scope, benefits
4. Application Architecture	System structure and design
5. Web Application Structure	Directory structure and deployment architecture
6. Servlet Development	Explanation of developed servlets
7. JSP Development	Explanation of JSP pages
8. Web Container Features	ServletContext, RequestDispatcher, Listeners
9. Session Management	Session implementation and tracking
10. Security Implementation	Authentication and authorisation
11. Deployment and Testing	Deployment process and test results
12. Operational Maintenance	Future maintenance and scalability
13. Conclusion	Summary of outcomes achieved
Appendices	Source Code, Screenshots, web.xml, Supporting Evidence

IMPLEMENTATION EVIDENCE

CATEGORY	EVIDENCE REQUIRED
Servlet Development	Servlet source code
JSP Development	JSP source code
Deployment Descriptor	web.xml
Session Management	Session implementation evidence
Listeners	Listener implementation
Deployment	Screenshots of deployed application
Testing	Functional testing evidence

SUBMISSION REQUIREMENTS

ITEM	FORMAT
Project Report	PDF
Source Code	ZIP
web.xml	Included in Source Code
Supporting Screenshots	Included in Report
Recommended Report Length	10–15 Pages (excluding appendices)

ASSESSMENT EVALUATION MATRIX

ASSESSMENT AREA	MARKS
Requirements Analysis and Application Planning	10
Web Application Structure and Deployment Design	10
Servlet Development	10
JSP Development	05
Web Container Features (Listeners, RequestDispatcher, ServletContext)	05
Session Management and Security	05
Deployment, Testing and Operational Maintenance	05
TOTAL	**50 MARKS**

COMPONENT B – RECORDED SOLUTION DEMONSTRATION

OVERVIEW

Students are required to prepare and submit an individual self-recorded video presentation explaining the web application developed for the assessment scenario.

The purpose of the recording is to demonstrate the student's understanding of the submitted solution and provide evidence of ownership prior to the Viva Voce Examination.

PRESENTATION REQUIREMENTS

Students must:

- Display their National Identity Card (NIC), Passport, Driving Licence, or other approved photographic identification at the beginning of the recording.
- Appear clearly on camera throughout the recording.
- Use their own voice throughout the presentation.
- Demonstrate personal understanding of the submitted solution.
- Clearly explain the work developed.

Medium: English

PRESENTATION CONTENT

Students should explain:

- Requirements Analysis
- Application Structure
- Servlet Development
- JSP Development
- Session Management
- RequestDispatcher Usage
- Listeners
- Security Implementation
- Deployment Process
- Key Technical Decisions

SUBMISSION FORMAT

- Video Format: MP4
- Duration: 5–10 Minutes

ASSESSMENT EVALUATION MATRIX

ASSESSMENT AREA	MARKS
Explanation of Requirements and Application Design	05
Explanation of Servlet and JSP Development	05
Explanation of Session Management, Security and Deployment	05
Communication, Professional Presentation, Authenticity and Ownership	05
TOTAL	20 MARKS

COMPONENT C – LIVE VIVA VOCE EXAMINATION AND PROJECT DEFENCE

OVERVIEW

Students are required to attend a Live Viva Voce Examination and Project Defence.

The purpose of the assessment is to verify:

- Knowledge and understanding.
- Practical Web Component Development skills.
- Application of web programming concepts.
- Problem-solving ability.
- Ownership and authenticity of submitted work.
- Ability to respond to new technical requirements.

DURATION

- Approximately 15–20 Minutes

VIVA VOCE REQUIREMENTS

Following the submission of Components A and B, students may request to undertake the Viva Voce Examination through the Student Portal before the published application deadline.

Students must be prepared to:

- Explain the submitted application.
- Explain servlet lifecycle and request processing.
- Explain deployment configuration.
- Explain JSP implementation.
- Explain RequestDispatcher usage.
- Explain listeners.
- Explain session management decisions.
- Explain security mechanisms.
- Justify technical decisions.
- Defend implementation approaches.

ASSESSMENT EVALUATION MATRIX

ASSESSMENT AREA	MARKS
Knowledge and Understanding	05
Application of Web Component Concepts	05
Analysis of Requirements and Technical Decisions	05
Evaluation, Justification and Defence	10
Ownership and Authenticity of Submitted Work	05
TOTAL	30 MARKS

APPLICATION FOR THE VIVA VOCE EXAMINATION

Students are responsible for applying for the Viva Voce Examination through the Student Portal before the published application deadline.

Failure to apply within the published deadline will result in the student being recorded as Absent and the assessment attempt being deemed unsuccessful.

The student will be required to undertake a repeat assessment, which may be subject to repeat assessment fees and a maximum achievable grade of Pass.

Students are therefore strongly advised to apply before the published application deadline.

ACADEMIC INTEGRITY AND AUTHENTICITY

Students are expected to maintain the highest standards of academic integrity throughout the assessment process.

All submitted reports, source code, deployment configurations, demonstrations, and Viva Voce responses must represent the student's own understanding and work.

Students may be required to participate in additional verification activities, demonstrations, live coding exercises, or oral questioning to verify authenticity, ownership, and academic integrity.

Failure to demonstrate reasonable ownership and understanding of submitted work may result in academic review procedures in accordance with institutional regulations.